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Manufacturing method of butadiene

Abstract

A manufacturing method of butadiene includes an electrolytic reduction process that produces ethylene and oxygen from carbon dioxide and water as a raw material by electrolytic reduction, a butene producing process that produces butene by dimerizing the ethylene produced in the electrolytic reduction process, a mixing process that prepares a mixed gas by mixing the oxygen produced in the electrolytic reduction process, the butene produced in the butene producing process, and air, and a butadiene producing process that produces butadiene by heating the mixed gas and oxidatively dehydrogenating the butene, wherein carbon dioxide by-produced in the butadiene producing process is used as a portion of the raw material in the electrolytic reduction process.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2005/0171311	12/2004	Schindler	526/335	C07C 5/3335
2014/0088331	12/2013	Rolland	585/326	B01J 31/0212
2020/0392632	12/2019	Oh et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2004176129	12/2003	JP	N/A
2011148720	12/2010	JP	N/A
2014062094	12/2013	JP	N/A
2015182985	12/2014	JP	N/A
2017080738	12/2016	JP	N/A
2017080739	12/2016	JP	N/A
2019085353	12/2018	JP	N/A

OTHER PUBLICATIONS

International Search Report for Japanese PCT Application No. PCT/JP2022/008576 mailed Apr. 26, 2022; 2 pp. cited by applicant

Extended European Search Report for European Patent Application No. 22766921.5 dated Feb. 18, 2025; 8 pp. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application is a U.S. National Phase Application of PCT/JP2022/008576, filed on Mar. 1, 2022, which claims the benefit of priority to Japanese Patent Application No. 2021-037409, filed Mar. 9, 2021. The contents of these applications are hereby expressly incorporated by reference in their entirety.

TECHNICAL FIELD

(2) The present invention relates to a manufacturing method of butadiene.

BACKGROUND ART

(3) In recent years, a manufacturing method of ethylene has shifted from a manufacturing method of pyrolyzing naphtha to a manufacturing method of using, as a raw material, ethane acquired from a shale gas and the like. In a case where ethane is used as a raw material, the amount of butadiene by-produced together with ethylene is reduced as compared with a case where naphtha is used as a raw material. Accordingly, there is a problem that a supply-demand gap of butadiene increases. To solve this problem, Patent Documents 1 and 2 disclose a method that produces butadiene by dimerizing ethylene to produce butene and then oxidatively dehydrogenating butene.

PRIOR ART DOCUMENT(S)

SUMMARY OF THE INVENTION

Task to be Accomplished by the Invention

(5) However, in a case where butadiene is produced by oxidatively dehydrogenating butene, there is a problem that a large amount of carbon dioxide is produced by a complete combustion reaction as a side reaction. Conventionally, the produced carbon dioxide is released into the atmosphere, but it is desirable to reduce the carbon dioxide from the viewpoint of global environmental conservation.

(6) In view of the above background, an object of the present invention is to provide a manufacturing method of butadiene that can reduce carbon dioxide emissions.

Means to Accomplish the Task

(7) To achieve such an object, one aspect of the present invention provides a manufacturing method of butadiene, comprising: an electrolytic reduction process (2) that produces ethylene and oxygen from carbon dioxide and water as a raw material by electrolytic reduction; a butene producing process (3) that produces butene by dimerizing the ethylene produced in the electrolytic reduction process; a mixing process (4) that prepares a mixed gas by mixing the oxygen produced in the electrolytic reduction process, the butene produced in the butene producing process, and air; and a butadiene producing process (5) that produces butadiene by heating the mixed gas and oxidatively dehydrogenating the butene, wherein carbon dioxide by-produced in the butadiene producing process is used as a portion of the raw material in the electrolytic reduction process.

(8) According to this aspect, ethylene as a raw material of butene and oxygen required for oxidative dehydrogenation can be produced by using carbon dioxide that is generated when butadiene is produced by oxidative dehydrogenation. Thus, the emissions of carbon dioxide as a greenhouse gas can be reduced. Further, the cost of a raw material can be reduced in the manufacturing method of butadiene.

(9) In the above aspect, preferably, the manufacturing method of butadiene comprises: a heat exchanging process (6) that cools a gas composition flowing out in the butadiene producing process and containing the butadiene and the carbon dioxide; a butadiene separating process (7) that separates the butadiene from the gas composition cooled in the heat exchanging process; and a carbon dioxide separating process (8) that separates the carbon dioxide from the gas composition from which the butadiene is separated in the butadiene separating process, wherein the carbon dioxide separated in the carbon dioxide separating process is used as the portion of the raw material in the electrolytic reduction process.

(10) According to this aspect, carbon dioxide is concentrated and supplied to the electrolytic reduction process. Accordingly, the efficiency of the electrolytic reduction can be improved.

(11) In the above aspect, preferably, in the heat exchanging process, the gas composition flowing out in the butadiene producing process and containing the butadiene and the carbon dioxide is cooled by exchanging heat with the gas composition from which the carbon dioxide is separated in the carbon dioxide separating process.

(12) According to this aspect, energy efficiency can be improved.

(13) In the above aspect, preferably, the gas composition from which the carbon dioxide is separated in the carbon dioxide separating process is mixed with the mixed gas in the mixing process after exchanging heat in the heat exchanging process with the gas composition flowing out in the butadiene producing process and containing the butadiene and the carbon dioxide.

(14) According to this aspect, the gas composition from which carbon dioxide is separated in the carbon dioxide separating process is supplied to the butadiene producing process via the mixing process after heated in the heat exchanging process. Accordingly, the energy consumption for heating a reactor of the butadiene producing process can be reduced.

(15) In the above aspect, preferably, the manufacturing method of butadiene comprises: measuring

an oxygen concentration and a flow rate of the mixed gas in the mixing process; and controlling a flow rate of the oxygen supplied from the electrolytic reduction process to the mixing process based on the oxygen concentration and the flow rate of the mixed gas.

(16) According to this aspect, the oxygen concentration of the mixed gas can be maintained within an appropriate range.

(17) In the above aspect, preferably, the manufacturing method of butadiene comprises: measuring a flow rate of the carbon dioxide supplied from the butadiene producing process to the electrolytic reduction process; and controlling a potential of the electrolytic reduction in the electrolytic reduction process based on the flow rate of the carbon dioxide.

(18) According to this aspect, the efficiency of the electrolytic reduction can be improved.

(19) In the above aspect, preferably, the manufacturing method of butadiene further comprises an ethylene producing process (**101**) that produces ethylene from ethane or naphtha as a raw material, wherein the ethylene produced in the ethylene producing process is used as a portion of a raw material in the butene producing process.

(20) According to this aspect, the supplying amount of ethylene can be increased.

(21) In the above aspect, preferably, butene by-produced in the ethylene producing process is used in the butadiene producing process.

(22) According to this aspect, the supplying amount of butene can be increased.

Effect of the Invention

(23) Thus, according to the above aspects, it is possible to provide a manufacturing method of butadiene that can reduce carbon dioxide emissions.

Description

BRIEF DESCRIPTION OF THE DRAWING(S)

(1) FIG. 1 is an explanatory diagram showing a butadiene manufacturing system;

(2) FIG. 2 is an explanatory diagram showing the butadiene manufacturing system;

(3) FIG. 3 is an explanatory diagram showing an electrolytic reductor; and

(4) FIG. 4 is an explanatory diagram showing a butadiene manufacturing system according to a modified embodiment.

MODE(S) FOR CARRYING OUT THE INVENTION

(5) In the following, an embodiment of a manufacturing method of butadiene according to the present invention will be described. As shown in FIGS. 1 and 2, a butadiene manufacturing system **1** according to the embodiment includes an electrolytic reduction process **2**, a butene producing process **3**, a mixing process **4**, a butadiene producing process **5**, a heat exchanging process **6**, a butadiene separating process **7**, and a carbon dioxide separating process **8**.

(6) In the electrolytic reduction process **2**, ethylene and oxygen are produced from carbon dioxide and water as a raw material by electrolytic reduction.



(7) An electrolytic reductor that uses a gas diffusion electrode for a cathode **16**, an electrolytic reductor that uses a solid polymer membrane as a separator, and the like may be used for the electrolytic reduction process **2**.

(8) As shown in FIG. 3, an electrolytic reductor **10** used in the electrolytic reduction process **2** may be, for example, a three-chamber electrolytic reductor. More specifically, the electrolytic reductor **10** may include an electrolytic cell **14** provided with a cathode gas chamber **11**, a catholyte chamber **12**, and an anolyte chamber **13** that are partitioned from each other. The cathode gas chamber **11** and the catholyte chamber **12** are partitioned by the cathode **16** as a gas diffusion electrode. The catholyte chamber **12** and the anolyte chamber **13** are partitioned by a partition wall **17** having ion conductivity. An anode **18** is arranged in the anolyte chamber **13**. Gaseous carbon

dioxide is supplied to the cathode gas chamber **11**. Carbon dioxide is supplied from the carbon dioxide separating process **8** as described later. Catholyte is supplied to the catholyte chamber **12**. Anolyte is supplied to the anolyte chamber **13**. The anode **18** and the cathode **16** are connected to a DC power supply **19**.

(9) The anolyte and the catholyte are aqueous solutions in which an electrolyte is dissolved. The electrolyte includes at least one of potassium, sodium, lithium, and a compound thereof. For example, the electrolyte may include at least one selected from the group consisting of LiOH, NaOH, KOH, Li_2CO_3 , Na_2CO_3 , K_2CO_3 , LiHCO_3 , NaHCO_3 , and KHCO_3 .

(10) The cathode **16** is a gas diffusion electrode, and includes a gas diffusion layer **21** and a microporous layer **22**. The gas diffusion layer **21** is permeable to a gas containing carbon dioxide, but inhibits the permeation of an aqueous solution containing the catholyte. The microporous layer **22** is permeable to both the gas containing carbon dioxide and the aqueous solution containing the catholyte. The gas diffusion layer **21** and the microporous layer **22** each have a flat surface. The gas diffusion layer **21** is arranged on a side of the cathode gas chamber **11**, and the microporous layer **22** is arranged on a side of the catholyte chamber **12**.

(11) For example, the gas diffusion layer **21** may be composed of a porous conductive base material such as carbon paper, carbon felt, and carbon cloth, and a water-repellent film such as polytetrafluoroethylene formed on a surface of the conductive base material. The conductive base material is connected to a negative pole of the DC power supply **19**, and supplied with electrons. The microporous layer **22** is formed on a surface of the gas diffusion layer **21** by using carbon black, and the like, and carries a catalyst. The catalyst may be a known carbon dioxide reduction catalyst, and includes, for example, at least one of a Group 11 element such as copper, a Group 12 element such as zinc, a Group 13 element such as gallium, a Group 14 element such as germanium, and a metal compound thereof. The metal compound includes at least one of an oxide, a sulfide, and a phosphide. The catalyst is preferably suitable for reducing carbon dioxide to produce ethylene, and is preferably made of a material formed by combining copper or a copper compound with a metal of a Group 11 element, a Group 12 element, a Group 13 element, or a Group 14 element or a metal compound thereof, for example. A binder such as an ion exchange resin may be added to the microporous layer **22**.

(12) For example, the anode **18** consists of a metal such as titanium, nickel, molybdenum, platinum, gold, silver, copper, iron, and lead, a metal alloy thereof, a carbon-based material such as carbon, or a conductive ceramic. The anode **18** may be formed into a flat plate, a flat plate with a plurality of openings, a mesh, or a porous body. The openings of the flat plate may be formed into a circle, a rhombus, a star, and the like. The flat plate may be corrugated or curved, and a surface thereof may be uneven. The anode **18** carries an oxygen generating catalyst such as platinum or iridium. The anode **18** may be provided on a surface of the partition wall **17** on a side of the anolyte chamber **13**.

(13) The DC power supply **19** converts the electric power, which is acquired by thermal power generation, nuclear power generation, solar power generation, wind power generation, hydroelectric power generation, and the like, into direct current as necessary, and supplies the electric power to the cathode **16** and the anode **18**. From the viewpoint of reducing carbon dioxide emissions, it is preferable to use, as the DC power supply **19**, the electric power acquired from solar power generation, wind power generation, hydroelectric power generation, and the like that use natural energy (renewable energy). The DC power supply **19** applies a voltage such that the cathode **16** has a negative potential with respect to the anode **18**. The DC power supply **19** acquires a potential of the cathode **16** by using a reference electrode, and preferably controls the voltage to be applied such that the potential of the cathode **16** is kept within a prescribed range.

(14) The cathode gas chamber **11** includes an inlet **24** and an outlet **25**. A carbon dioxide gas is supplied from the inlet **24** and discharged from the outlet **25**. The outlet **25** of the cathode gas

chamber **11** is connected to the inlet **24** via a gas circulation passage **26**.

(15) The catholyte chamber **12** includes an inlet **27** and an outlet **28**. The inlet **27** and the outlet **28** of the catholyte chamber **12** are connected via a catholyte circulation passage **29**. Similarly, the anolyte chamber **13** includes an inlet **31** and an outlet **32**. The inlet **31** and the outlet **32** of the anolyte chamber **13** are connected via an anolyte circulation passage **33**. The catholyte circulation passage **29** and the anolyte circulation passage **33** are provided with separators **35** and **36**, respectively. The separators **35** and **36** may include a gas-liquid separator. Further, the catholyte circulation passage **29** and the anolyte circulation passage **33** may be provided with electrolyte concentration controllers **37** and **38**, respectively, for adjusting electrolyte concentrations of the catholyte and the anolyte within a prescribed range. The electrolyte concentration controllers **37** and **38** may include a sensor that detects the electrolyte concentrations of the catholyte and the anolyte, an electrolyte supply device that supplies fresh catholyte and anolyte at a prescribed concentration, and a drainage device that discharges a portion of circulating catholyte and anolyte.

(16) Further, a gas circulation flow rate adjuster **34** that discharges a portion of a gas circulating inside the gas circulation passage **26** is provided in the gas circulation passage **26**. An outlet of the gas circulation flow rate adjuster **34** is connected to a first gas passage **39**. A gas discharge passage of the separator **35** is connected to the first gas passage **39**. The gas circulation flow rate adjuster **34** adjusts a flow rate and a pressure of a gas circulating through the gas circulation passage **26** and the cathode gas chamber **11** by discharging the gas to the first gas passage **39**. A gas pressure in the cathode gas chamber **11** is maintained by the gas circulation flow rate adjuster **34** so as to be higher by a prescribed value than a liquid pressure in the catholyte chamber **12**. Accordingly, the catholyte in the catholyte chamber **12** is inhibited from passing through the cathode **16** and flowing into the cathode gas chamber **11**. A portion of the gas in the cathode gas chamber **11** passes through the cathode **16**, and flows into the catholyte chamber **12**. It is preferable that the amount of gas flowing from the cathode gas chamber **11** into the catholyte chamber **12** is small.

(17) Carbon dioxide in the cathode gas chamber **11** diffuses inside the gas diffusion layer **21** of the cathode **16** and is reduced in the microporous layer **22**, and thus a first product is acquired. The first product contains ethylene as a main product and a by-product such as methane, hydrogen, carbon monoxide, and formic acid. A large portion of the first product is generated at the cathode **16** on the side of the cathode gas chamber **11**. A portion of the first product is generated at the cathode **16** on the side of the catholyte chamber **12**. The first product in the catholyte chamber **12** is mixed with unreacted carbon dioxide that flows into the catholyte chamber **12**. Similarly, the first product in the cathode gas chamber **11** is mixed with unreacted carbon dioxide.

(18) Among the components of the first product generated at the cathode **16** on the side of the catholyte chamber **12**, ethylene, methane, and by-produced hydrogen and carbon monoxide are gases, and are separated from the catholyte together with unreacted carbon dioxide by the separator **35** of the catholyte circulation passage **29**, and then flow into the first gas passage **39**. The first gas passage **39** may be provided with a separator that separates ethylene from the first product. The separator may be configured by combining a distillation device, an extraction device, and an adsorption device.

(19) Among the components of the first product generated at the cathode **16** on the side of the cathode gas chamber **11**, ethylene, methane, and by-produced hydrogen and carbon monoxide circulate through the gas circulation passage **26** together with unreacted carbon dioxide, and are discharged from the gas circulation flow rate adjuster **34** into the first gas passage **39**. The first product, which includes unreacted carbon dioxide as well as ethylene, methane, and by-produced hydrogen and carbon monoxide generated at the cathode **16** on the side of the catholyte chamber **12**, flows from the separator **35** and the gas circulation flow rate adjuster **34** into the first gas passage **39**.

(20) At the anode **18**, water and hydroxide ions in the anolyte are oxidized, and thus gaseous oxygen is generated. Oxygen is separated from the anolyte by the separator **36** of the anolyte

circulation passage **33**. At the anode **18**, water and hydroxide ions in the anolyte are oxidized, and thus oxygen is generated. Oxygen is a gas, and is separated from the anolyte by the separator **36** of the anolyte circulation passage **33** and flows into a second gas passage **40**.

(21) In the electrolytic reduction process **2**, potentials of the cathode **16** and the catalyst carried by the cathode **16** may be set such that Faraday efficiency for producing ethylene at the cathode **16** is 30% or more, preferably 50% or more. Faraday efficiency is defined as a ratio of the current that contributes to the production of each product to the total current that flows through the electrolytic cell **14**. In the electrolytic reduction process **2**, the catalyst carried by the cathode **16** may be selected such that the selectivity for producing ethylene at the cathode **16** is 30% or more.

(22) The ethylene produced in the electrolytic reduction process **2** is supplied from the first gas passage **39** to the butene producing process **3** via a first line **41**. Further, the oxygen produced by the electrolytic reduction process **2** is supplied from the second gas passage **40** to the mixing process **4** via a second line **42**.

(23) In the butene producing process **3**, ethylene produced in the electrolytic reduction process **2** is dimerized, and thus butene is produced.

2C.sub.2H.sub.4.fwdarw.C.sub.4H.sub.8

(24) In the butene producing process **3**, butene is produced, and a main product thereof is n-butene. As shown in FIG. **2**, the butene producing process **3** includes a dimerization reactor **44** and a first separator **45**.

(25) The dimerization reactor **44** may be, for example, a fixed bed flow reactor filled with an ethylene dimerization catalyst. The ethylene dimerization catalyst contains nickel, alumina, and silica. The ethylene dimerization catalyst may be, for example, a catalyst in which alumina and nickel are carried by a silica carrier, or a catalyst in which nickel is carried by a silica carrier containing alumina. The nickel content of the ethylene dimerization catalyst is 0.0001% to 1% by weight, preferably 0.0001% to 0.5% by weight, more preferably 0.0001% to 0.05% by weight.

(26) The carrier preferably has a high specific surface area and a high pore volume. The specific surface area of the carrier is preferably 200 m²/g to 1200 m²/g, and the pore volume of the carrier is preferably 0.4 cc/g to 2 cc/g. The silica carrier may be made of amorphous silica or mesoporous silica. The carrier containing silica and alumina may be made of Y-type zeolite, X-type zeolite, mordenite, beta-type zeolite, L-type zeolite, or MFI-type zeolite.

(27) In the dimerization reactor **44**, a reaction temperature of a dimerization reaction of ethylene is set to 150° C. to 400° C., preferably 200° C. to 350° C. In a case where the reaction temperature is lower than 150° C., the activity of the catalyst is lowered. In a case where the reaction temperature is higher than 400° C., branched olefin rapidly increases, nickel tends to aggregate on the catalyst, and coke tends to be by-produced. Accordingly, the catalyst may be deteriorated. A reaction pressure of dimerization of ethylene is preferably 0.1 MPa to 50 MPa. In a case where the reaction pressure is higher than 50 MPa, a by-product tends to be produced. In a case where the reaction pressure is lower than 0.1 MPa, the activity of the catalyst is lowered. The feeding velocity (weight hourly space velocity: WHSV) of ethylene per unit weight of the catalyst may be 0.1 h⁻¹ to 50 h⁻¹, preferably 0.5 h⁻¹ to 40 h⁻¹, and more preferably 0.5 h⁻¹ to 30 h⁻¹. In a case where the feeding velocity of ethylene is less than 0.1 h⁻¹, the productivity is lowered. Further, the selectivity of a dimer and a trimer is lowered as successive reactions of oligomerization progress. In a case where the feeding velocity of ethylene is greater than 40 h⁻¹, a conversion rate of ethylene is lowered.

(28) N-butene, which includes 1-butene, cis-2-butene, and trans-2-butene, is produced as a main product by the dimerization reaction of ethylene. Further, hexene such as 1-hexene, 2-hexene, and 3-hexene may be produced as a by-product.

(29) The first separator **45** separates n-butene from unreacted ethylene and a reaction product acquired by the dimerization of ethylene in the dimerization reactor **44**.

(30) The first separator **45** is connected to the dimerization reactor **44** via a third line **47**. The first

separator **45** may be configured by combining a known distillation device, extraction device, and adsorption device. Further, the first separator **45** may separate unreacted ethylene from the reaction product, and return unreacted ethylene to the dimerization reactor **44** via a return line **48**.

Hydrocarbon such as hexene separated by the first separator **45** may be transmitted to an after-mentioned oxygen combustion device **60** and used as fuel.

(31) In the mixing process **4**, oxygen produced in the electrolytic reduction process **2**, butene produced in the butene producing process **3**, and air are mixed, and thus a mixed gas is prepared. The mixing process **4** includes a gas mixer **51**. The gas mixer **51** is supplied with n-butene from the first separator **45** of the butene producing process **3** via a fourth line **52**, with air via an air line **53**, with oxygen from the separator **36** of the electrolytic reduction process **2** via the second line **42**, and with recycled gas from the carbon dioxide separating process **8** via a sixth line **55**. The recycled gas mainly contains nitrogen and oxygen. Air, oxygen from the electrolytic reduction process **2**, and the recycled gas are used to adjust an oxygen concentration in an after-mentioned oxidative dehydrogenation reactor **67**. The mixed gas is prepared such that a molar ratio of oxygen:n-butene is kept in the range of 1:0.5 to 1:3, preferably in the range of 1:0.8 to 1:2.

(32) The air line **53** is provided with a first flow rate control valve **57** that controls a flow rate of air supplied to the gas mixer **51**. The second line **42** is provided with a second flow rate control valve **58** that controls a flow rate of oxygen supplied to the gas mixer **51**. The second flow rate control valve **58** is connected via a seventh line **59** to the oxygen combustion device **60** such as a boiler that uses oxygen. Thermal energy generated in the oxygen combustion device **60** may be recovered and used to heat a gas flowing out of the gas mixer **51**. Further, carbon dioxide produced by combustion in the oxygen combustion device **60** may be recovered and used as a portion of a raw material in the electrolytic reduction process **2**.

(33) The gas mixer **51** supplies the mixed gas containing ethylene, oxygen, air, and the recycled gas to the butadiene producing process **5** via an eighth line **61**. An outlet of the gas mixer **51** or the eighth line **61** is provided with a first gas flowmeter **63** that measures a flow rate of the mixed gas, and an oxygen concentration meter **64** that measures an oxygen concentration of the mixed gas. Further, a pressure pump **65** that pressurizes the mixed gas and a furnace **66** that preheats the mixed gas are provided on the eighth line **61**. The furnace **66** may be supplied with oxygen from the second line **42** or the seventh line **59**.

(34) In the butadiene producing process **5**, the mixed gas is heated and butene is oxidatively dehydrogenated, and thus butadiene is produced.

$C_{4}H_{8} + \frac{1}{2}O_{2} \rightarrow C_{4}H_{6} + H_{2}O$

(35) The butadiene producing process **5** includes the oxidative dehydrogenation reactor **67**. The oxidative dehydrogenation reactor **67** may be any reactor such as a fixed bed reactor, an ebullient bed reactor, and a moving bed reactor. The oxidative dehydrogenation reactor **67** is filled with an oxidative dehydrogenation catalyst. The oxidative dehydrogenation catalyst may be a composite metal oxide catalyst containing molybdenum and bismuth, an iron oxide catalyst, a vanadium oxide catalyst, and the like. The oxidative dehydrogenation catalyst preferably contains iron and cobalt in addition to molybdenum and bismuth. The oxidative dehydrogenation catalyst may contain silica in addition to composite metal oxide.

(36) In the oxidative dehydrogenation reactor **67**, an oxidative dehydrogenation reaction is carried out at 300° C. to 600° C., preferably 300° C. to 500° C., more preferably 320° C. to 460° C. Further, the oxidative dehydrogenation reaction is carried out at 0 MPa to 2 MPa, preferably 0 MPa to 0.5 MPa. Further, a feeding velocity of n-butene per unit weight of the catalyst may be 0.1 h.⁻¹ to 10 h.⁻¹, more preferably 0.2 h.⁻¹ to 5 h.⁻¹. Butadiene as a main product is produced from n-butene by the oxidative dehydrogenation reaction. Further, carbon dioxide as a by-product is produced by complete combustion of n-butene.

(37) An outlet of the oxidative dehydrogenation reactor **67** is connected to the butadiene separating process **7** via a ninth line **69**. From the outlet of the oxidative dehydrogenation reactor **67**, a gas

composition containing butadiene, carbon dioxide, and unreacted mixed gas flows out. The ninth line **69** is provided with a heat exchanger **71** that constitutes the heat exchanging process **6**. The gas composition passing through the ninth line **69** is cooled in the heat exchanger **71**.

(38) The gas composition is supplied to the butadiene separating process **7** via the ninth line **69**, and butadiene is separated from the gas composition in the butadiene separating process **7**. The butadiene separating process **7** includes a second separator **72**. The second separator **72** may, for example, liquefy butadiene by cooling the gas composition, and separate liquid butadiene from the gas composition by gas-liquid separation.

(39) The gas composition from which butadiene is separated in the butadiene separating process **7** mainly contains nitrogen, carbon dioxide, and oxygen. The gas composition from which butadiene is separated is supplied to the carbon dioxide separating process **8** via a tenth line **74**, and carbon dioxide is separated from the gas composition in the carbon dioxide separating process **8**. The carbon dioxide separating process **8** may include a third separator **75** that works by using a known process such as a chemisorption process such as a Benfield process and an MDEA (methyldiethanolamine) process, a physical adsorption process such as a Selexol process and a Rectisol process, a membrane separation process, a PSA process (pressure swing adsorption process), a PTSA process (pressure temperature swing adsorption process), an electrochemical separation process that uses quinone, and the like. Carbon dioxide separated from the gas composition and adsorbed by various adsorbents is separated from the adsorbents by a regeneration process, and turns into a gaseous state with a high concentration.

(40) Carbon dioxide separated in the carbon dioxide separating process **8** is supplied from the third separator **75** to the cathode gas chamber **11** of the electrolytic reduction process **2** via an eleventh line **76**. Accordingly, the carbon dioxide by-produced in the butadiene producing process **5** is used as a portion of a raw material in the electrolytic reduction process **2**. Further, the eleventh line **76** is connected to a carbon dioxide line **77** that supplies a carbon dioxide gas to the eleventh line **76**. The carbon dioxide line **77** is provided with a third flow rate control valve **78**. The eleventh line **76** is provided with a carbon dioxide concentration meter **79** that measures a concentration of carbon dioxide passing through the eleventh line **76**. Further, the eleventh line **76** is provided with a second gas flowmeter **80** that measures a flow rate of carbon dioxide passing through the eleventh line **76**.

(41) The gas composition from which carbon dioxide is separated mainly contains nitrogen and oxygen, and is returned as the recycled gas to the gas mixer **51** of the mixing process **4** via the sixth line **55**. The sixth line **55** passes through the heat exchanger **71** of the heat exchanging process **6**, and the recycled gas exchanges heat in the heat exchanger **71** with the gas composition passing through the ninth line **69**. Accordingly, at an outlet of the heat exchanger **71**, the temperature of the recycled gas passing through the sixth line **55** increases, while the temperature of the gas composition passing through the ninth line **69** decreases.

(42) A controller **85** controls the first flow rate control valve **57** and the second flow rate control valve **58** based on signals from the first gas flowmeter **63** and the oxygen concentration meter **64**. The controller **85** may increase, based on the signal from the first gas flowmeter **63**, the opening degree of the first flow rate control valve **57** as the flow rate of the mixed gas decreases. Accordingly, the amount of air supplied to the gas mixer **51** increases, and thus the flow rate of the mixed gas increases. Further, the controller **85** may adjust, based on the signal from the oxygen concentration meter **64**, the opening degree of the second flow rate control valve **58** and increase the flow rate of oxygen flowing from the second flow rate control valve **58** to the gas mixer **51** as the oxygen concentration of the mixed gas decreases. Accordingly, the oxygen concentration of the mixed gas can be maintained within an appropriate range.

(43) Further, the controller **85** may control the potential of the DC power supply **19** based on signals from the carbon dioxide concentration meter **79** and the second gas flowmeter **80**. Accordingly, the efficiency of the electrolytic reduction is improved. Further, the controller **85**

controls the third flow rate control valve **78** based on the signals from the carbon dioxide concentration meter **79** and the second gas flowmeter **80**, and adjusts the amount of carbon dioxide supplied to the carbon dioxide separating process **8**.

(44) The effect of the above embodiment will be described. The butadiene manufacturing system **1** and the manufacturing method of butadiene can produce ethylene as a raw material of butene and oxygen required for oxidative dehydrogenation by using carbon dioxide that is generated when butadiene is produced by oxidative dehydrogenation. Thus, the emission of carbon dioxide as a greenhouse gas generated during the production of butadiene can be reduced. Further, the cost of a raw material can be reduced in the manufacturing method of butadiene.

(45) Carbon dioxide generated in the butadiene producing process **5** is concentrated through the heat exchanging process **6**, the butadiene separating process **7**, and the carbon dioxide separating process **8**, and is supplied to the electrolytic reduction process **2**. Accordingly, the efficiency of the electrolytic reduction can be improved. The gas composition from which the carbon dioxide is separated in the carbon dioxide separating process **8** is mixed with the mixed gas in the mixing process **4** after exchanging heat in the heat exchanging process **6** with the gas composition flowing out in the butadiene producing process **5** and containing the butadiene and the carbon dioxide. Accordingly, the energy consumption for heating the oxidative dehydrogenation reactor **67** of the butadiene producing process **5** can be reduced. Further, the gas composition flowing out in the butadiene producing process **5** and containing butadiene and carbon dioxide can be cooled, and thus energy efficiency can be improved.

(46) As shown in FIG. **4**, the butadiene manufacturing system **1** may further include an ethylene producing process **101** that produces ethylene from ethane or naphtha as a raw material. The ethylene producing process **101** may produce ethylene by ethane cracking of ethane as a raw material or naphtha cracking of naphtha as a raw material. A portion of ethylene produced in the ethylene producing process **101** may be supplied to the butene producing process **3** via a twelfth line **102**, and used as a portion of a raw material in the butene producing process **3**.

(47) Further, butene by-produced in the ethylene producing process **101** may be supplied to the mixing process **4** via a thirteenth line **103** and used in the butadiene producing process **5**. In this case, the twelfth line **102** may be omitted. Further, in a case where the ethylene producing process **101** is provided, the carbon dioxide line **77** may be omitted since the amount of ethylene that needs to be produced in the electrolytic reduction process **2** is reduced.

(48) As an example, in the three-chamber electrolytic cell **14** composed of the cathode gas chamber **11**, the catholyte chamber **12**, and the anolyte chamber **13**, the cathode **16** was formed of a gas diffusion electrode that carries a copper-zinc composite catalyst, and the anode **18** was formed of a Pt mesh. The electricity was applied at 265 mA/cm² for 6 hours while a potassium hydrogen carbonate aqueous solution of 1M (1 mol/L) was supplied at 1 mL/min to each of the catholyte chamber **12** and the anolyte chamber **13**, and carbon dioxide was supplied at 100 mL/min to the cathode gas chamber **11**. Analysis of the products in the gas showed that the cathode **16** produced 37% ethylene, 1% methane, and 25% hydrogen in Faraday efficiency, and the anode **18** produced a gas with 99% oxygen in the Faraday efficiency.

(49) In the butadiene manufacturing system shown in FIG. **4**, a process calculation was performed in a case where 100 kg/hr of butadiene was produced. In a case where 100 kg/hr of butadiene was produced in the butadiene producing process **5**, 12 kg/hr of ethylene was supplied from the electrolytic reduction process **2** to the butene producing process **3**, 250 kg/hr of ethylene was supplied from the ethylene producing process **101** to the butene producing process **3**, 205 kg/hr of butene was supplied from the butene producing process **3** to the mixing process **4**, air containing 10 kg/hr of oxygen was supplied from the air line **53** to the mixing process **4**, 56 kg/hr of oxygen was supplied from the electrolytic reduction process **2** to the mixing process **4**, and 50 kg/hr of carbon dioxide was supplied from the butadiene producing process **5** to the electrolytic reduction process **2**. That is, 50 kg/hr of carbon dioxide was produced in the butadiene producing process **5**, 12 kg/hr

of ethylene and 56 kg/hr of oxygen were produced from this carbon dioxide as a raw material in the electrolytic reduction process **2**. In this case, the selectivity of ethylene in the electrolytic reduction process **2** was set at 80%.

(50) Concrete embodiments of the present invention have been described in the foregoing, but the present invention should not be limited by the foregoing embodiments and various modifications and alterations are possible within the scope of the present invention.

GLOSSARY OF TERMS

(51) **1**: butadiene manufacturing system **2**: electrolytic reduction process **5**: butadiene producing process **6**: heat exchanging process **7**: butadiene separating process **8**: carbon dioxide separating process **10**: electrolytic reductor **44**: dimerization reactor **51**: gas mixer **57**: first flow rate control valve **58**: second flow rate control valve **61**: oxygen combustion device **63**: first gas flowmeter **64**: oxygen concentration meter **67**: oxidative dehydrogenation reactor **71**: heat exchanger **72**: second separator **85**: controller **101**: ethylene producing process **102**: twelve line **103**: thirteenth line

Claims

1. A manufacturing method of butadiene, comprising: an electrolytic reduction process that produces ethylene and oxygen from carbon dioxide and water as a raw material by electrolytic reduction; a butene producing process that produces butene by dimerizing the ethylene produced in the electrolytic reduction process; a mixing process that prepares a mixed gas by mixing the oxygen produced in the electrolytic reduction process, the butene produced in the butene producing process, and air; and a butadiene producing process that produces butadiene by heating the mixed gas and oxidatively dehydrogenating the butene, wherein carbon dioxide by-produced in the butadiene producing process is used as a portion of the raw material in the electrolytic reduction process.
2. The manufacturing method of butadiene according to claim 1, comprising: a heat exchanging process that cools a gas composition flowing out in the butadiene producing process and containing the butadiene and the carbon dioxide; a butadiene separating process that separates the butadiene from the gas composition cooled in the heat exchanging process; and a carbon dioxide separating process that separates the carbon dioxide from the gas composition from which the butadiene is separated in the butadiene separating process, wherein the carbon dioxide separated in the carbon dioxide separating process is used as the portion of the raw material in the electrolytic reduction process.
3. The manufacturing method of butadiene according to claim 2, wherein in the heat exchanging process, the gas composition flowing out in the butadiene producing process and containing the butadiene and the carbon dioxide is cooled by exchanging heat with the gas composition from which the carbon dioxide is separated in the carbon dioxide separating process.
4. The manufacturing method of butadiene according to claim 3, wherein the gas composition from which the carbon dioxide is separated in the carbon dioxide separating process is mixed with the mixed gas in the mixing process after exchanging heat in the heat exchanging process with the gas composition flowing out in the butadiene producing process and containing the butadiene and the carbon dioxide.
5. The manufacturing method of butadiene according to claim 1, comprising: measuring an oxygen concentration and a flow rate of the mixed gas in the mixing process; and controlling a flow rate of the oxygen supplied from the electrolytic reduction process to the mixing process based on the oxygen concentration and the flow rate of the mixed gas.
6. The manufacturing method of butadiene according to claim 1, comprising: measuring a flow rate of the carbon dioxide supplied from the butadiene producing process to the electrolytic reduction process; and controlling a potential of the electrolytic reduction in the electrolytic reduction process based on the flow rate of the carbon dioxide.

7. The manufacturing method of butadiene according to claim 1, further comprising an ethylene producing process that produces ethylene from ethane or naphtha as a raw material, wherein the ethylene produced in the ethylene producing process is used as a portion of a raw material in the butene producing process.

8. The manufacturing method of butadiene according to claim 7, wherein butene by-produced in the ethylene producing process is used in the butadiene producing process.
